C10.1 About external snoop filtering

The ACE protocol supports a mechanism for constructing an optional external snoop filter that operates with an existing cached master component.

Note

External snoop filtering is optional. If a master component supports external snoop filtering, it must declare this in its data sheet.

To support the addition of a snoop filter, a cached master must ensure that it broadcasts sufficient information to permit a snoop filter to track shareable allocations and evictions for cache lines that the master maintains. This ensures that a snoop filter can:

- reduce the number of snoop transactions required to be passed to the master, thus reducing cache intrusion
- in certain circumstances, provide a faster response to snoop transactions.

For correct operation, a snoop filter must observe any transactions being issued by a master, that could result in that master allocating a cache line in its local cache. The snoop filter must also observe activity that indicates an eviction from the local cache of that master. This includes:

- local cache line evictions
- WriteBack of cache lines to memory
- snoop transactions that cause an eviction.

Whether a transaction causes a cache line to be allocated depends on the transaction. A snoop filter can determine the expected allocation state of a particular cache line by observing the transactions issued by a master component. Table C10-1 shows the expected allocation state for both cache maintenance transactions and normal transactions. If the actual allocation for a cache line is different from what the filter expects, then an associated Evict operation must be performed, to ensure that the snoop filter can correctly track the allocated cache line.

Table C10-1 External snoop filter expected cache line allocation states

Transaction	Expected cache line allocation
ReadOnce	Allocation does not change
ReadClean	Allocated
ReadNotSharedDirty	Allocated
ReadShared	Allocated
ReadUnique	Allocated
CleanUnique	Allocated
MakeUnique	Allocated
CleanShared	Allocation does not change
CleanInvalid	Evicted
MakeInvalid	Evicted
WriteUnique	Allocation does not change
WriteLineUnique	Allocation does not change
WriteClean	Allocation does not change
WriteBack	Evicted
Evict	Evicted

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An error response to a transaction does not change the allocation behavior of a snoop filter. Therefore, any master that changes its allocation behavior when a transaction receives an error response must take appropriate action to ensure the snoop filter is kept up to date.

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